Jesse Ames
Student ID: 2424350

AMATH 583 Spring 2025 Final

1.) Graph a

It does have a Hamiltonian cycle...

a → b → h → g → k → i → c → d → j → f → e → a

Graph b

It does have a Hamilton cycle

a → c → h → f → i → j → g → e → b → d → a

Graph c

It does have a hamiltonian cycle.

a → h → e → f → g → i → d → c → b → a

Graph D

Has neither a Hamilton path (& therefore) nor a cycle. Due to sections isolated by having to pass through vertex e, so would have to pass through twice.

Graph E

Has a Hamilton path, but not a cycle.

a → b → c → d → e → j → i → h → g → f → k → l → m → n → o

Graph F

It does have a hamilton cycle...

a → f → k → p → q → l → g → h → m → r → s → t → o → n → i → j → e → d → c → b → a

Problem 2.

$\forall x$ on the real line, find all real-valued continously differentiable functions f. s.t.

$$(f(x))^2 = 2025 + \int_0^x ((f(t))^2 + (f'(t))^2)\, dt$$

- differentiate both sides d/dx

$$2f(x)f'(x) = (f(x))^2 + (f'(x))^2$$

$$(f(x))^2 + (f'(x))^2 - 2f(x)f'(x) = 0$$

$\rightarrow$ Perfect square: $(f'(x) - f(x))^2 = 0$

$$f'(x) - f(x) = 0 \rightarrow f'(x) = f(x)$$

$\rightarrow$ general ODE $\rightarrow$ $f(x) = f(0)\, e^x$

we can then sub into initial equation to get initial condition

$$(f(0))^2 = 2025 + \int_0^0 (f(t))^2 + (f'(t))^2\, dt$$

$$f(0)^2 = 2025 \rightarrow f(0) = \sqrt{2025}$$

$$f(0) = \pm 45$$

so the functions are $\boxed{f(x) = \pm 45 e^x}$